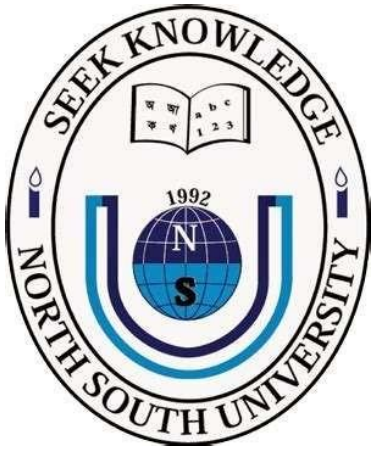
Analog Electronics II: Op-amp & 555 Timer



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Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 10

Unit 3: Filters Reference

1. Operarional Amplifiers and Linear Integrated Circuits by Robert Coughlin- Chapter-11

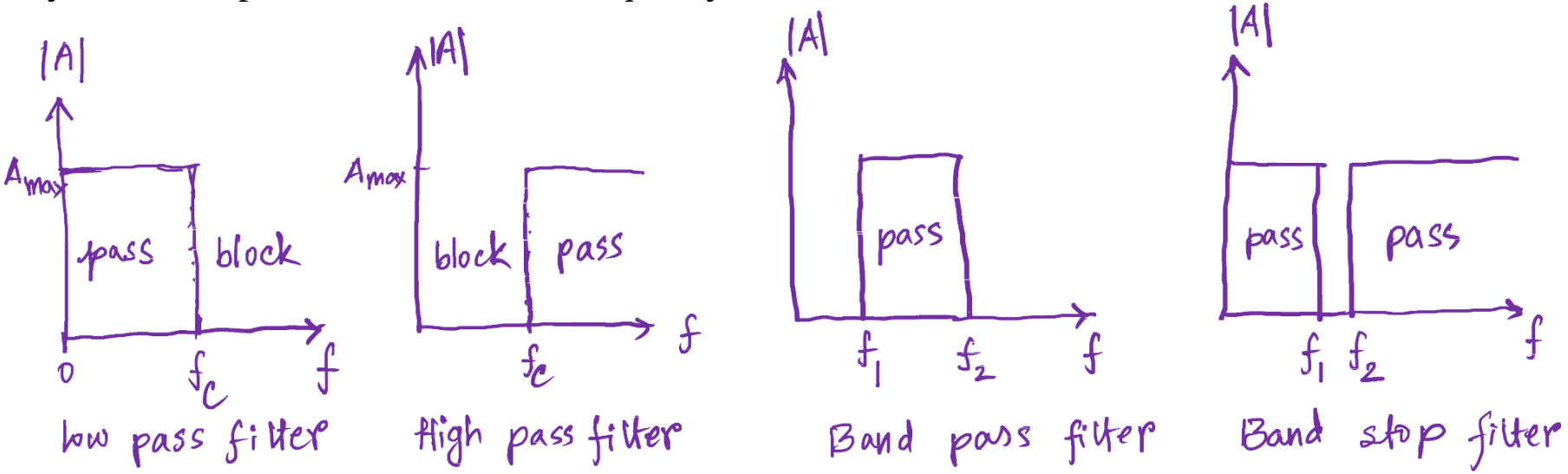
2. Fundamentals of Electric Circuits by Alexander Sadiku- Chapter-14

Frequency

The **frequency response** of a circuit is the variation in its behavior with change in signal frequency.

The sinusoidal steady-state frequency responses of circuits are of significance in many applications, especially in communications and control systems.

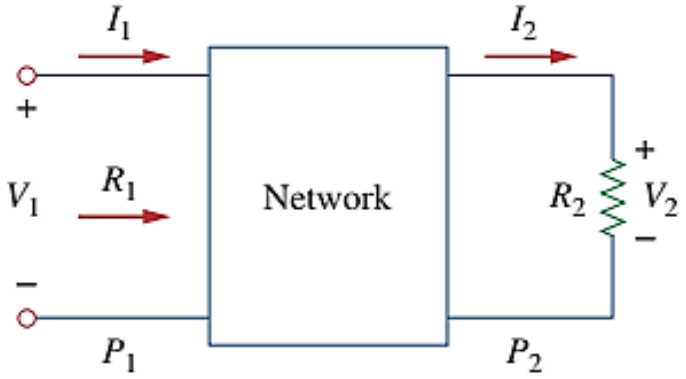
A specific application is in electric filters that block out or eliminate signals with unwanted frequencies and pass signals of the desired frequencies. Filters are used in radio, TV, and telephone systems to separate one broadcast frequency from another.



Source: Alexander-Sadiku Chapter 14

Decibel

Alternatively, the gain G can be expressed in terms of voltage or current ratio.



$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{V_2^2/R_2}{V_1^2/R_1}$$

$$= 10 \log_{10} \left(\frac{V_2}{V_1} \right)^2 + 10 \log_{10} \frac{R_1}{R_2}$$

$$G_{\text{dB}} = 20 \log_{10} \frac{V_2}{V_1} - 10 \log_{10} \frac{R_2}{R_1}$$

For the case when $R_2 = R_1$, $\quad G_{\text{dB}} = 20 \log_{10} \dfrac{V_2}{V_1}$

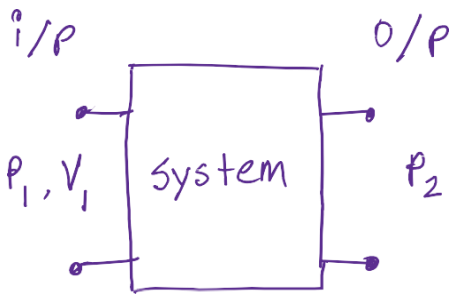
if $P_1 = I_1^2 R_1$ and $P_2 = I_2^2 R_2$,

for $R_1 = R_2$,

$$G_{\text{dB}} = 20 \log_{10} \frac{I_2}{I_1}$$

Source: Alexander-Sadiku Chapter 14

Decibel



P_1 = input power

P_2 = output "

V_1 = input voltage

V_2 = output "

The *decibel* (dB) provides us with a unit of less magnitude. It is 1/10th of a bel and is given by

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} \tag{14.5}$$

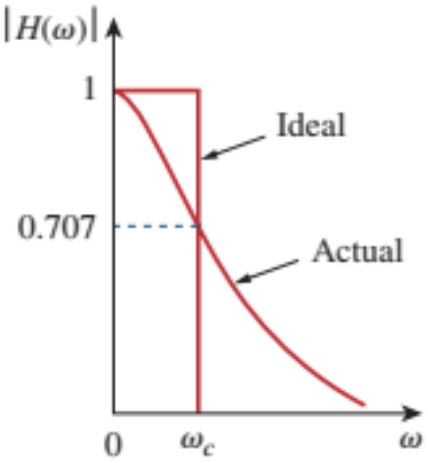
When $P_1 = P_2$, there is no change in power and the gain is 0 dB. If $P_2 = 2P_1$, the gain is

$$G_{\text{dB}} = 10 \log_{10} 2 \simeq 3 \text{ dB} \tag{14.6}$$

and when $P_2 = 0.5P_1$, the gain is

$$G_{\text{dB}} = 10 \log_{10} 0.5 \simeq -3 \text{ dB} \tag{14.7}$$

The half-power frequency, which is equivalent to the corner frequency on the Bode plots in the context of filters is usually known as the cutoff frequency is obtained by setting the magnitude of equal to $1/\sqrt{2}$

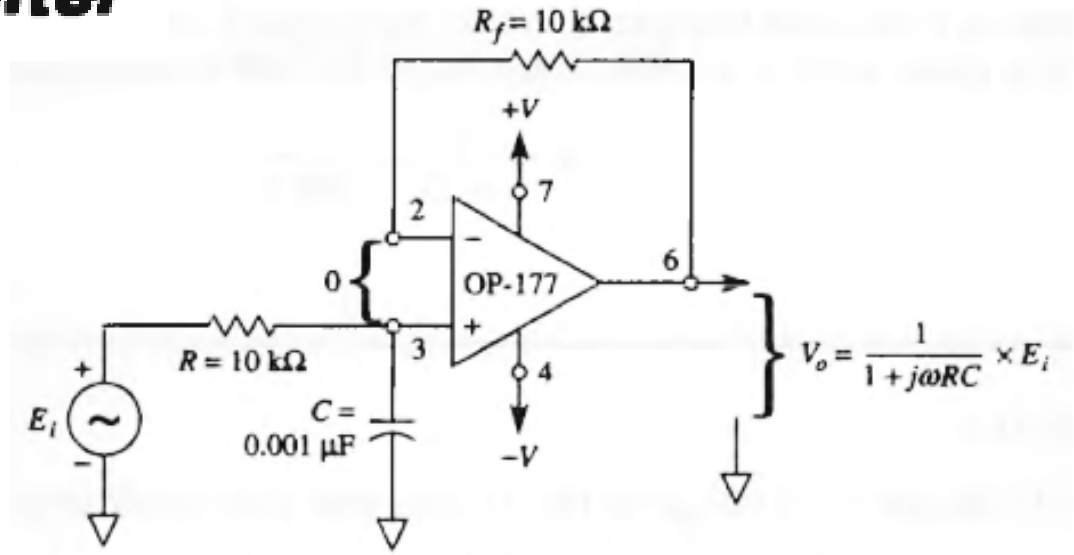


Source: Alexander-Sadiku Chapter 14

Active Filters: First-Order Lowpass Filter

BUTTERWORTH FILTER

– 20-dB/Decade Filter



(a) Low-pass filter for a roll-off of −20 dB/decade.

Designing the Filter

$$V_o = \frac{1/j\omega C}{R + 1/j\omega C} \times E_i$$

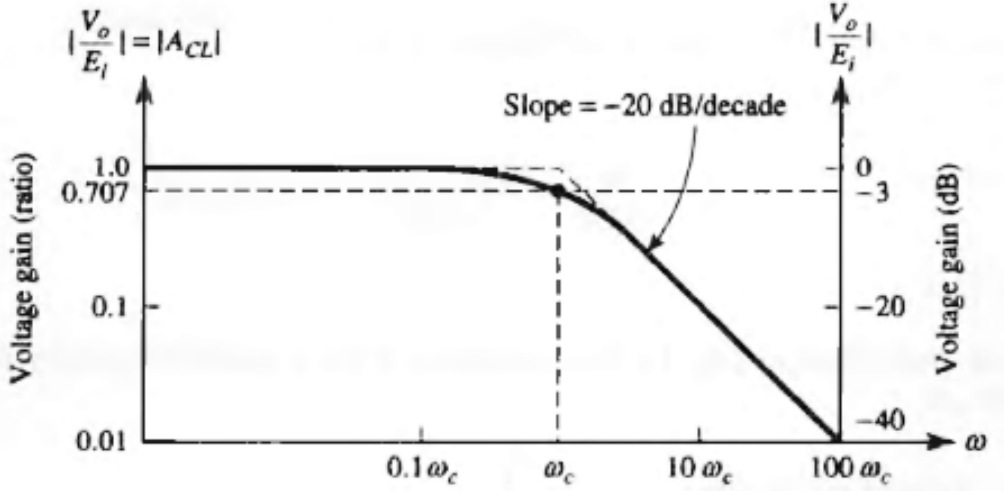
$$A_{CL} = \frac{V_o}{E_i} = \frac{1}{1 + j\omega RC}$$

$$\omega_c = \frac{1}{RC} = 2\pi f_c$$

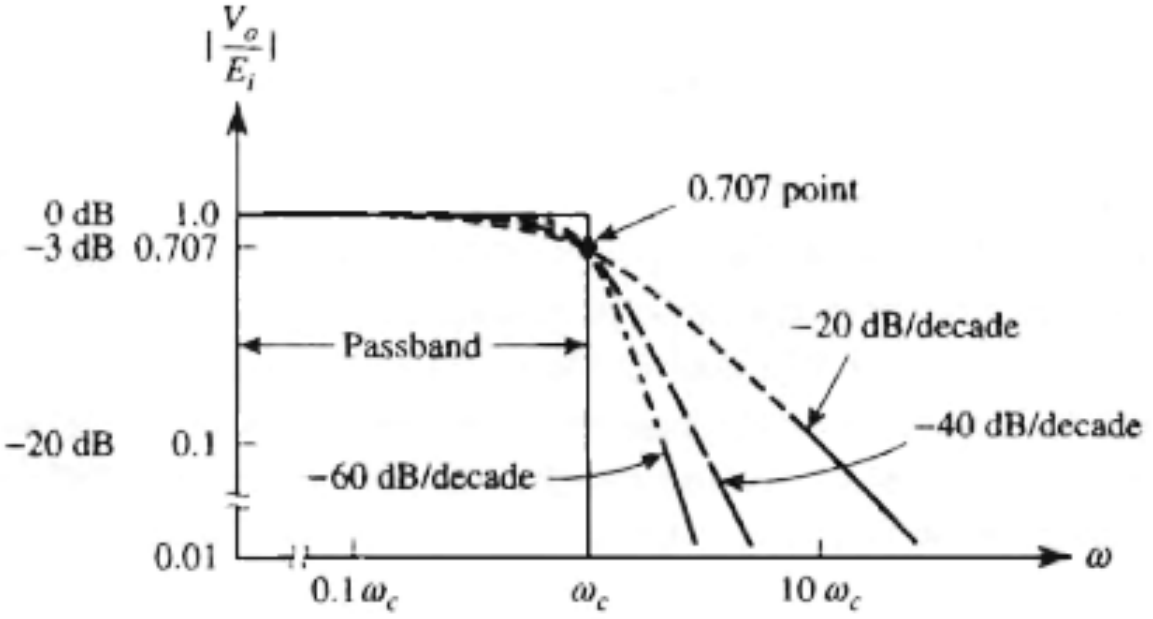
Active Low Pass Filter Frequency Response

$$\text{dB} = 20 \log_{10} \frac{Vo}{E_i}$$

BASIC LOW-PASS FILTER



BUTTERWORTH FILTER



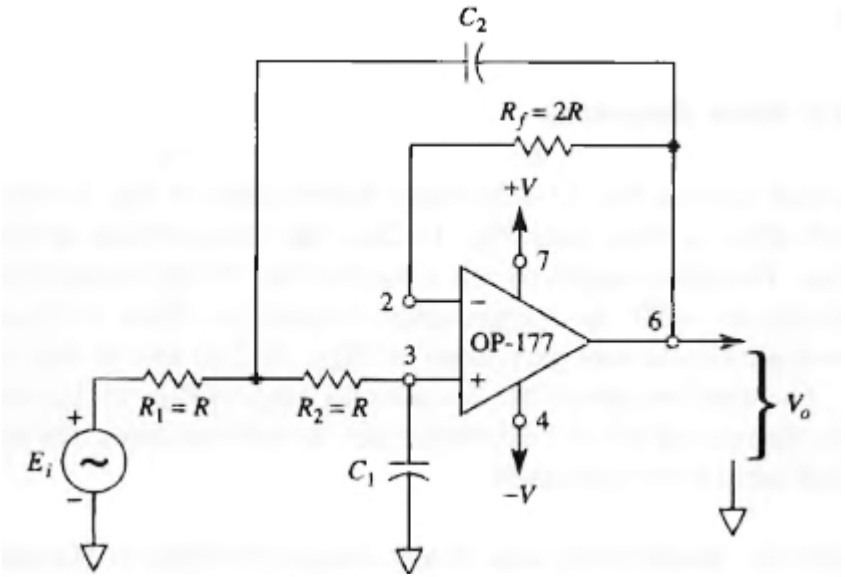
— 40-DB/DECADE LOW-PASS BUTTERWORTH FILTER

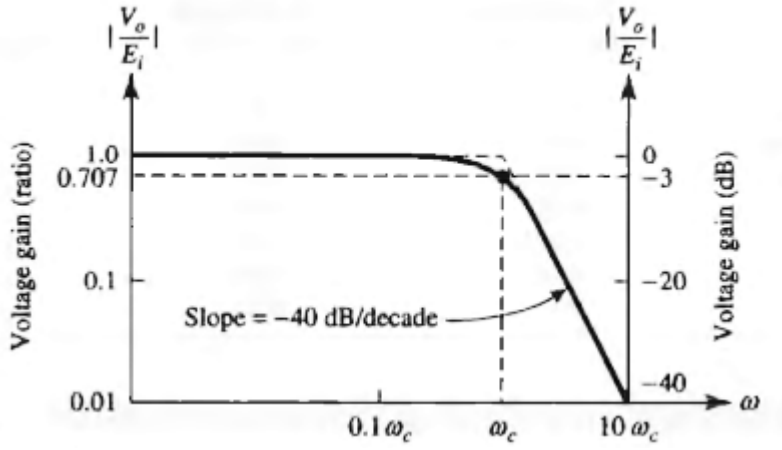
Design procedure

1. Choose the cutoff frequency, ω_c or f_c.
2. Pick C_1; choose a convenient value between 100 pF and 0.1 μF.
3. Make $C_2 = 2C_1$.
4. Calculate

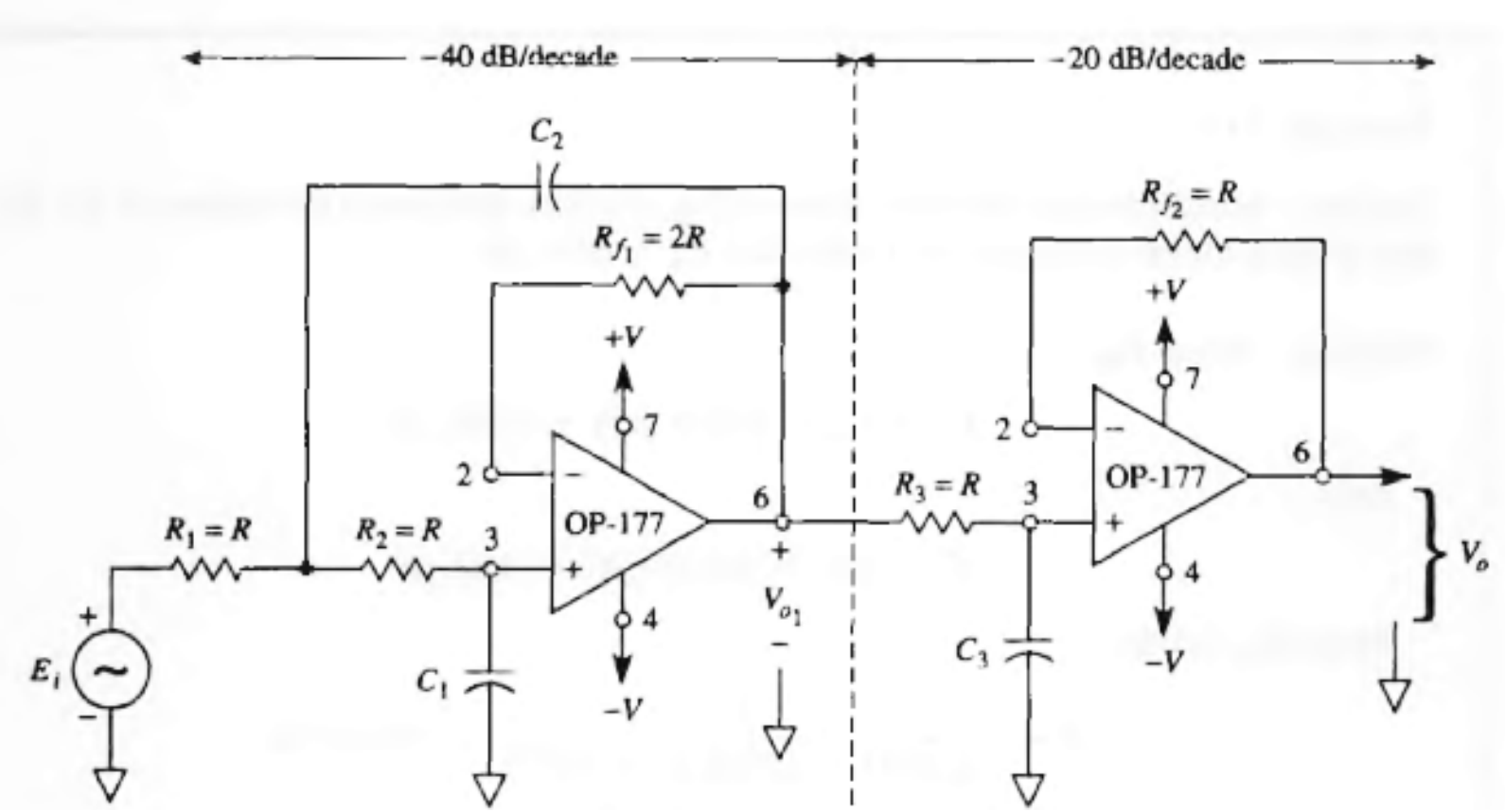
$$R = \frac{0.707}{\omega_c C_1}$$

5. Choose $R_f = 2R$.





– 60-DB/DECADE LOW-PASS BUTTERWORTH FILTER



— 60-DB/DECADE LOW-PASS BUTTERWORTH FILTER

Design procedure

1. Choose the cutoff frequency, ω_c or f_c.
2. Pick C_3; choose a convenient value between 0.001 and 0.1 μF.
3. Make

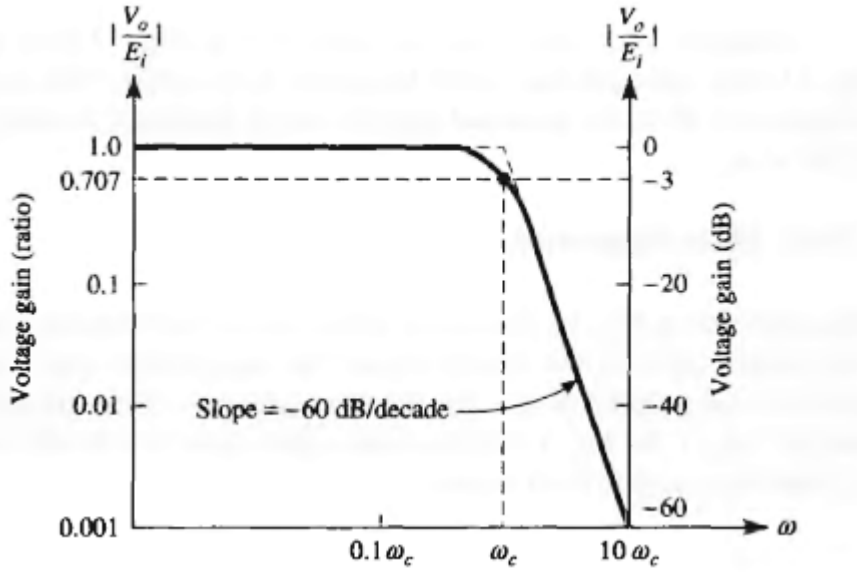
$$C_1 = \tfrac{1}{2}C_3 \qquad \text{and} \qquad C_2 = 2C_3$$

4. Calculate

$$R = \frac{1}{\omega_c C_3}$$

5. Make $R_1 = R_2 = R_3 = R$.
6. $R_{f_1} = 2R$ and $R_{f_2} = R$.

For best results the value of R should be between 10 and 100 kΩ.



$|A_{CL}|$ FOR THE LOW-PASS FILTERS

ω	−20 dB/decade; Fig. 11-2(a)	−40 dB/decade; Fig. 11-4(a)	−60 dB/decade; Fig. 11-5(a)
$0.1\omega_c$	1.0	1.0	1.0
$0.25\omega_c$	0.97	0.998	0.999
$0.5\omega_c$	0.89	0.97	0.992
ω_c	0.707	0.707	0.707
$2\omega_c$	0.445	0.24	0.124
$4\omega_c$	0.25	0.053	0.022
$10\omega_c$	0.1	0.01	0.001

Example 11-4

Determine R_1 and R_2 in Fig. 11-4(a) for a cutoff frequency of 1 kHz. Let $C_1 = 0.01\ \mu F$.

Solution Pick $C_2 = 2C_1 = 2(0.01\ \mu F) = 0.02\ \mu F$. Select $R_1 = R_2 = R$ from Eq. (11.3):

$$R = \frac{0.707}{(6.28)(1 \times 10^3)(0.01 \times 10^{-6})} = 11{,}258\ \Omega$$

and

$$R_f = 2(11{,}258\ \Omega) = 22{,}516\ \Omega$$

Example 11-5

For the -60-dB/decade low-pass filter of Fig. 11-5(a), determine the values of C_1, C_2, and R for a cutoff frequency of 1 kHz. Let $C_3 = 0.01\ \mu F$.

Solution From Eq. (11-5),

$$C_1 = \tfrac{1}{2}C_3 = \tfrac{1}{2}(0.01\ \mu F) = 0.005\ \mu F$$

and

$$C_2 = 2C_3 = 2(0.01\ \mu F) = 0.02\ \mu F$$

From Eq. (11-6),

$$R = \frac{1}{(6.28)(1 \times 10^3)(0.01 \times 10^{-6})} = 15{,}915\ \Omega$$

Example Problem 14.12

Design a lowpass active filter with a dc gain of 4 and a corner frequency of 500 Hz.

Solution:

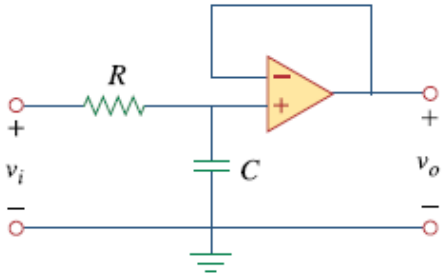
The corner frequency is

$$\omega_c = 2\pi f_c = 2\pi(500) = \frac{1}{R_f C_f}$$

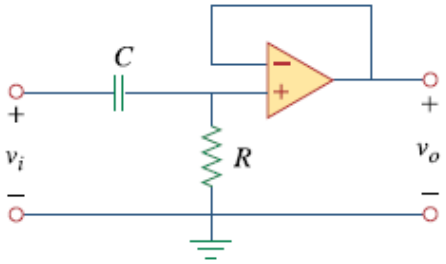
If we select $C_f = 0.2\ \mu\text{F}$ $\qquad R_f = \dfrac{1}{2\pi(500)0.2 \times 10^{-6}} = 1.59\ \text{k}\Omega$

Practice Problem

Find the transfer function for each of the active filters



(a)



(b)

Solution: (a) $\mathbf{V}_+ = \dfrac{1/j\omega C}{R + 1/j\omega C}\mathbf{V}_i, \qquad \mathbf{V}_- = \mathbf{V}_o$

Since $\mathbf{V}_+ = \mathbf{V}_-$,

$$\frac{1}{1 + j\omega RC}\mathbf{V}_i = \mathbf{V}_o$$

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \underline{\frac{\mathbf{1}}{\mathbf{1 + j\omega RC}}}$$

(b) $\mathbf{V}_+ = \dfrac{R}{R + 1/j\omega C}\mathbf{V}_i, \qquad \mathbf{V}_- = \mathbf{V}_o$

Since $\mathbf{V}_+ = \mathbf{V}_-$,

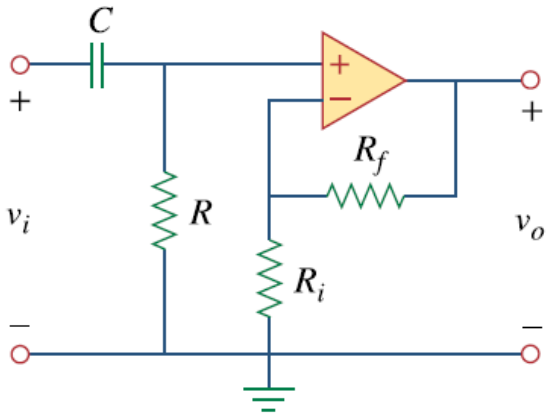
$$\frac{j\omega RC}{1 + j\omega RC}\mathbf{V}_i = \mathbf{V}_o$$

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \underline{\frac{\mathbf{j\omega RC}}{\mathbf{1 + j\omega RC}}}$$

Practice Problem

A highpass filter is shown in Fig. 14.92. Show that the transfer function is

$$\mathbf{H}(\omega) = \left(1 + \frac{R_f}{R_i}\right) \frac{j\omega RC}{1 + j\omega RC}$$



Solution:

$$\mathbf{V}_+ = \frac{\mathrm{R}}{\mathrm{R} + 1/\mathrm{j}\omega\mathrm{C}}\mathbf{V}_\mathrm{i} = \frac{\mathrm{j}\omega\mathrm{RC}}{1 + \mathrm{j}\omega\mathrm{RC}}\mathbf{V}_\mathrm{i}$$

$$\mathbf{V}_- = \frac{\mathrm{R_i}}{\mathrm{R_i} + \mathrm{R_f}}\mathbf{V}_\mathrm{o}$$

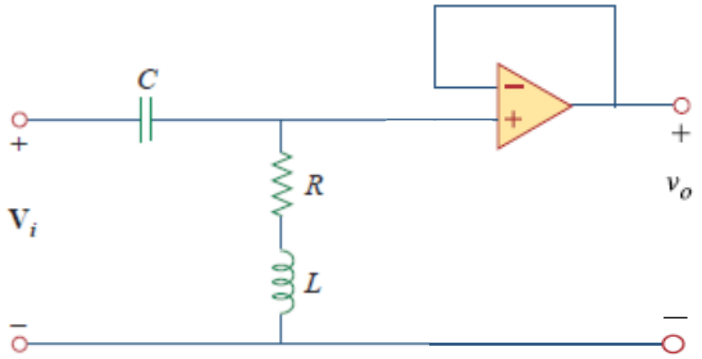
Since $\mathbf{V}_+ = \mathbf{V}_-$,

$$\frac{\mathrm{R_i}}{\mathrm{R_i} + \mathrm{R_f}}\mathbf{V}_\mathrm{o} = \frac{\mathrm{j}\omega\mathrm{RC}}{1 + \mathrm{j}\omega\mathrm{RC}}\mathbf{V}_\mathrm{i}$$

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_\mathrm{o}}{\mathbf{V}_\mathrm{i}} = \underline{\left(1 + \frac{\mathbf{R_f}}{\mathbf{R_i}}\right)\left(\frac{\mathbf{j}\omega\mathbf{RC}}{\mathbf{1} + \mathbf{j}\omega\mathbf{RC}}\right)}$$

Practice Problem

Find the transfer function $H(\omega)= V_0/V_i$ of the circuit below. What kind of filter it is and why? Sketch the approximate frequency response.



Solution:

Practice Problem

Design a low-pass filter for a biomedical signal conditioning circuit used to measure ECG signals. The filter must effectively remove high-frequency noise (e.g., muscle artifacts and power line interference) while preserving the ECG waveform.

Design Requirements:
Desired cutoff frequency, fc = 150 Hz
Roll-off rate: –60 dB/decade
Passband gain: 1 (unity gain)
Input Resistance: at least 5kΩ

Solution: